Relationship of dietary saturated fatty acids and body habitus to serum insulin concentrations: the Normative Aging Study.

The purpose of this study was to examine the relationship of body mass index, abdomen-hip ratio, and dietary intake to fasting and postprandial insulin concentrations among 652 men aged 43-85 y, followed in the Normative Aging Study. Log-transformed fasting insulin was significantly associated with body mass index, abdomen-hip ratio, total fat energy, and saturated fatty acid energy, with correlation coefficients ranging from 0.14 for total fat to 0.45 for body mass index. When multivariate models were used, body mass index, abdomen-hip ratio, and saturated fatty acid intake were statistically significant independent predictors of both fasting and postprandial insulin concentrations, after age, cigarette smoking, and physical activity were adjusted for. If saturated fatty acids as a percentage of total energy were to decrease from 14% to 8%, there would be an 18% decrease in fasting insulin and a 25% decrease in postprandial insulin. These data suggest that overall adiposity, abdominal obesity, and a diet high in saturated fatty acids are independent predictors for both fasting and postprandial insulin concentrations.